



Recitation 12

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1. Homework

2. Review

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2. A motorist goes through 4 lights, each of which is found to be red with probability $1/2$. The waiting times at each light are modeled as independent normal random variables with mean 1 minute and standard deviation $1/2$ minute. Let X be the total waiting time at the red lights.

(a) Use the total probability theorem to find the PDF and the transform associated with X , and the probability that X exceeds 4 minutes. Is X normal?



Using the total probability theorem, we have

$$f_X(x) = \sum_{k=0}^4 \mathbf{P}(k \text{ lights are red}) f_X(x | k \text{ lights are red})$$

With

$$\mathbf{P}(k \text{ lights are red}) = \binom{4}{k} \left(\frac{1}{2}\right)^4$$

The conditional PDF of X given that k lights are red is the sum of k normal RV with mean 1 and variance $1/4$, so it is also normal, with mean k and variance $k/4$.

And note that for $k = 0$ case, $\mathbf{P}(X = 0 | 0 \text{ lights are red}) = 1$.
Therefore,

$$f_X(x | 0 \text{ lights are red}) = \delta(x)$$

For $\mathbf{P}(X > 4)$, we have

$$\mathbf{P}(X > 4) = \sum_{k=0}^4 \mathbf{P}(k \text{ lights are red}) \mathbf{P}(X > 4 | k \text{ lights are red}).$$

$\mathbf{P}(X > 4 | k \text{ lights are red})$ can be calculated from the normal table.

X is not normal for either reason below:

1. X is mixture (linear combination) of Gaussian.
2. The transform of X is not in the form of normal RV.

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- Some useful inequalities
 - **Markov inequality**

$$\mathbf{P}(X \geq a) \leq \frac{\mathbf{E}[X]}{a}, \quad \text{for all } a > 0.$$

- **Chebyshev's inequality**

$$\mathbf{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

- **Chernoff inequality**

$$\mathbf{P}(X \geq a) \leq e^{-\max_{s \geq 0} (sa - \ln M_X(s))}.$$

- Weak law of large numbers

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- Convergence “in probability”

$$\lim_{n \rightarrow \infty} \mathbf{P}(|Y_n - a| \geq \epsilon) = 0$$

Upper bound for variance

- When X is known to take values in a range $[a, b]$, we claim that $\sigma^2 \leq (b - a)^2/4$.

- Note that for any constant γ , we have

$$\mathbf{E}[(X - \gamma)^2] = \mathbf{E}[X^2] - 2\mathbf{E}[X]\gamma + \gamma^2.$$

- The above quadratic is minimized when $\gamma = \mathbf{E}[X]$. It follows that

$$\sigma^2 = \mathbf{E}[(X - \mathbf{E}[X])^2] \leq \mathbf{E}[(X - \gamma)^2], \quad \text{for all } \gamma.$$

- By letting $\gamma = (a + b)/2$, we obtain

$$\sigma^2 \leq \mathbf{E}\left[\left(X - \frac{a+b}{2}\right)^2\right] = \mathbf{E}[(X-a)(X-b)] + \frac{(b-a)^2}{4} \leq \frac{(b-a)^2}{4}.$$

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1. RV X is uniformly distributed between -1.0 and 1.0 . Let $X_1, X_2, \dots$ be independent identically distributed RVs with the same distribution as X . Determine which, if any, of the following sequences (all with $i = 1, 2, \dots$) are convergent in probability.

(a) $U_i = \frac{X_1 + X_2 + \dots + X_i}{i}$

(b) $W_i = \max\{X_1, \dots, X_i\}$

(c) $V_i = X_1 \cdot X_2 \cdot \dots \cdot X_i$

Solution:

- (a) According to the weak law of large numbers,

$$\mathbf{P}(|U_i - 0| \geq \epsilon) \leq \frac{1}{3i\epsilon^2}.$$

As $i \rightarrow \infty$, $U_i \rightarrow 0$.

- (b) The value of W_i will not decrease as i increases. Thus, we have

$$\begin{aligned}\mathbf{P}(|W_i - 1| \leq \epsilon) &= 1 - \mathbf{P}(|W_i - 1| \geq \epsilon) \\ &= 1 - \mathbf{P}(1 - X_1 \geq \epsilon) \cdots \mathbf{P}(1 - X_i \geq \epsilon) \\ &= 1 - \left(1 - \frac{\epsilon}{2}\right)^i.\end{aligned}$$

In particular, $\lim_{i \rightarrow \infty} \mathbf{P}(|W_i - 1| \leq \epsilon) = 1 - \lim_{i \rightarrow \infty} \left(1 - \frac{\epsilon}{2}\right)^i \rightarrow 1$. Thus, we have $W_i \rightarrow 1$ as $i \rightarrow \infty$.

(c) $\min\{|X_1|, |X_2|, \dots, |X_i|\}$ converges to 0 in probability,

$$\begin{aligned} & \lim_{i \rightarrow \infty} \mathbf{P}(|\min\{|X_1|, |X_2|, \dots, |X_i|\} - 0| > \epsilon) \\ &= \lim_{i \rightarrow \infty} \mathbf{P}(|X_1| > \epsilon) \mathbf{P}(|X_2| > \epsilon) \cdots \mathbf{P}(|X_i| > \epsilon) \\ &= \lim_{i \rightarrow \infty} (1 - \epsilon)^i \\ &= 0. \end{aligned}$$

Since $|V_i| \leq \min\{|X_1|, |X_2|, \dots, |X_i|\}$ and $|V_i| \geq 0$, $|V_i|$ converges to 0 in probability.

2. Demonstrate that the Chebyshev inequality is tight, that is, for every μ , $\sigma > 0$, and $c \geq \sigma$, construct a RV X with mean μ and standard deviation σ such that

$$\mathbf{P}(|X - \mu| \geq c) = \frac{\sigma^2}{c^2} .$$

Solution:

Consider a RV X with PMF

$$p_X(x) = \begin{cases} p, & \text{if } x = \mu - c \\ p, & \text{if } x = \mu + c \\ 1 - 2p, & \text{if } x = \mu. \end{cases}$$

Thus, $\mathbf{E}[X] = \mu$ and $\text{var}(X) = 2pc^2$. By setting $p = \frac{\sigma^2}{2c^2}$, we have

$$\mathbf{P}(|X - \mu| \geq c) = \frac{2pc^2}{c^2} = \frac{\sigma^2}{c^2},$$

and therefore the Chebyshev inequality is tight.